

Total No. of Questions : 8]

SEAT No. :

PD4112

[Total No. of Pages : 2

[6402]-72

S.E. (Artificial Intelligence and Machine Learning)

COMPUTER GRAPHICS

(2019 Pattern) (Semester - IV) (218555)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Use the Cohen Sutherland Line Clipping Algorithm with the help of region codes to clip a line AB with A (30, 70), B (110, 50) and PQ with P(60, 120), Q (90, 30) to clip a line against a window with lower left-hand corner (40, 40) and Upper right-hand corner (100, 80). Show Graphic Representation of Original and Clipped Line. **[9]**

- b) Explain the basic transformation techniques in 3D Graphics. **[9]**
- i) Scaling
 - ii) Rotation
 - iii) Reflection about XZ Plane

OR

Q2) a) Let ABCD be the rectangle window with A (150, 150), B (150, 200), C(200, 200) and D (200, 150). Use Cohen Hodgeman polygon clipping algorithm to clip the convex polygon PQRS with P (100, 175), Q(170, 250), R(250, 165), S(180, 100) and find the final coordinates of the clipped polygon. **[9]**

- b) What is projection? Explain with diagram, oblique - Cavalier, Cabinet, Orthographic - isometric, diametric, trimetric Parallel projections. **[9]**

Q3) a) Define Shading. Compare Constant Intensity, Halftoning, Gourand Shading and Phong Shading algorithm. **[9]**

- b) Explain in detail with Diagram **[8]**
- i) RGB Color Model.
 - ii) HSV Color Model.
 - iii) CIE Chromaticity Diagram.
 - iv) Color Gamut.

OR

P.T.O.

Q4) a) What is a segment? Why do we need segments? Explain the complete process of [9]

i) Segment Creation,

ii) Segment Renaming and

iii) Segment Closing.

b) Define Illumination. Explain with diagram Phong illumination Model and Combined Diffuse Illumination models in detail. [8]

Q5) a) Write short note on Hilbert's and Koch Curve along its Topological and Fractal Dimensions. [9]

b) What are the steps in design in animation sequence? Describe about each step briefly. [9]

OR

Q6) a) Write short note on : [9]

i) Design of animation sequence

ii) Frame-by-frame Animation techniques

b) What is curve interpolation? As far as splines are concerned what do Bezier and B-splines curves indicate? [9]

Q7) a) What is the different usage of Virtual Reality? Explain in detail. [6]

b) What is Haptics Rendering Pipeline Modeling in Virtual Reality? [6]

c) Explain 3D position trackers. [5]

OR

Q8) a) What is graphics rendering pipeline in a Virtual Reality system. [6]

b) Explain gesture interfaces in Virtual Reality. [6]

c) What is kinematic modeling in a Virtual Reality? [5]

